API 610 10TH EDITION (ISO 13709)
CENTRIFUGAL PUMP PROCESS DATA SHEET
(SI UNITS)

MR No. MR95-G-8-1 | Item No. 95G-3215 | Rev.
Inquiry No. MR95-G-8-1
By: C.Yu | Date: 31-Jan-08

APPLICABLE TO: ○ PROPOSAL ● PURCHASE ◎ AS BUILT
FOR Suncor Energy | UNIT Steam Generation
SITE Firebag Stage 5, Fort McMurray, Alberta | SERVICE LP BFW Pump
NOTES: INFORMATION BELOW TO BE COMPLETED: ○ BY PURCHASER ☐ BY MANUFACTURER ◎ BY MANUFACTURER OR PURCHASER

DATA SHEETS (6.1.1)

	ITEM NO.(S) / DATA SHEET NO.(S)	ATTACHED
PUMP	95G-3215	●
MOTOR	95GM-3215	●
GEAR		○
TURBINE		○

APPLICABLE STANDARD(S): Firebag Project standard FB-G-5616

Pump Information

Manufacturer	Flowserve. Desio, Italy
Model	UZDL
Size	6 UZDL 25
No. of Stages	2
Serial No.'s	D66893 1B

● OPERATING CONDITIONS (5.1.3) @ SG = 0.97

CAPACITY, NORMAL 545 (m^3 / h) RATED 654 (m^3 / h)
OTHER
SUCTION PRESS. MAX. / RATED 143.4 / 20 (kPag)
DISCHARGE PRESSURE 2,763 (kPag)
DIFFERENTIAL PRESSURE 2,743 (kPa)
DIFF. HEAD 288.3 (m) NPSHA 5.96 (m)
PROCESS VARIATIONS (5.1.4)
STARTING CONDITIONS (5.1.4)
SERVICE: ● CONT. ○ INTERMITTENT (STARTS / DAY)
● PARALLEL OPERATION REQ'D (5.1.13) HYD. POWER 498 (kW)

● SITE DATA (5.1.3)

LOCATION: (5.1.30)
● INDOOR ● HEATED ○ OUTDOOR ○ UNHEATED
● ELECTRICAL AREA CLASSIFICATION (5.1.24 / 6.1.4) UNCLASSIFIED
CL. GR. Zone
○ WINTERIZATION REQ'D ○ TROPICALIZATION REQ'D.
SITE DATA (5.1.30)
● ALTITUDE 593.4 (m) BAROMETER 94.4 kPa (Abs.)
● RANGE OF AMBIENT TEMP, MIN. / MAX. + 5 / + 40 (°C)
○ RELATIVE HUMIDITY: MIN. / MAX. / (%)
UNUSUAL CONDITIONS: (5.1.30) ○ DUST ○ FUMES
○ OTHER
Refer to Site Conditions Data Sheet DS93-A-1-1 for more information.

● DRIVER TYPE

● INDUCTION MOTOR ○ STEAM TURBINE ○ GEAR
○ OTHER

● MOTOR DRIVER (6.1.1 / 6.1.4)

● MANUFACTURER TECO - Westinghouse
■ (1000 HP) 745.7 (kW) ■ 1800 (RPM)
■ FRAME 4010 ● ENCLOSURE WPII
● HORIZONTAL ◎ VERTICAL ● SERVICE FACTOR 1.15
● VOLTAGE / PH. / HZ. 4000 / 3 / 60
● TYPE Squirrel Cage Induction
● MINIMUM STARTING VOLTAGE (6.1.5)
● INSULATION Class F ● TEMP. RISE Class B
● FULL LOAD AMPS
● LOCKED ROTOR AMPS
● STARTING METHOD Across the line
● LUBE See Note 2

BEARINGS (TYPE / NUMBER):
■ RADIAL split - sleeve /
■ THRUST split - sleeve /
☐ VERTICAL THRUST CAPACITY
UP (N) DOWN (N)
See Motor Data Sheet DS95-P-GM3215-1.

● LIQUID (5.1.3)

LIQUID TYPE OR NAME Boiler Feed Water
○ HAZARDOUS ○ FLAMMABLE ○ (5.1.5)

	MIN.	NORMAL	MAX.
PUMPING TEMP (°C)	10	85	98
VAPOR PRESS. kPa (Abs.)	1.2	57.8	94.3
RELATIVE DENSITY (SG):	1	0.97	0.96
VISCOSITY (mPa.s)	1.3	0.3301	0.2849

SPECIFIC HEAT, Cp 4.2 @ 85°C (kJ / kg °K)
● CHLORIDE CONCENTRATION 3000 (ppmw)
○ H_2S CONCENTRATION (6.5.2.4) (PPMW) WET (5.2.1.12c)
CORROSIVE / EROSIVE AGENT TDS < 8000 ppmw (5.12.1.9)

MATERIALS

● ANNEX H CLASS (5.12.1.1) S-6
● MIN. DESIGN METAL TEMP. (5.12.4.1) -29 (°C)
● REDUCED HARDNESS MATERIALS REQ'D. (5.12.1.12)
■ BARREL / CASE A216WCB IMPELLER A487CA6NM
■ CASE / IMPELLER WEAR RINGS A473-420
■ SHAFT A276-410 1B
■ DIFFUSERS

● PERFORMANCE: @ 1.00 SG

PROPOSAL CURVE NO. 6UZDL25-BS1-F ■ 1783 RPM 1B
■ IMPELLER DIA. RATED 593 MAX. 650.0 MIN. 500.0 (mm) 1B
■ IMPELLER TYPE CLOSED
■ RATED POWER 687.4 (kW) EFFICIENCY 72.5 (%) 1B
■ MINIMUM CONTINUOUS FLOW:
THERMAL (m^3 / h) STABLE 300 (m^3 / h)
■ PREFERRED OPER. REGION 438 TO 750 (m^3 / h)
■ ALLOWABLE OPER. REGION 300 TO 750 (m^3 / h)
■ MAX HEAD @ RATED IMPELLER 342 (m) 1B
■ MAX POWER @ RATED IMPELLER 745 (kW) 1B
■ NPSHR AT RATED FLOW 4.0 (m) (5.1.10) 1B
● MAX. SUCTION SPECIFIC SPEED 9204 (5.1.11)
● MAX. SOUND PRESS. LEVEL REQ'D. <85 (dBA) (5.1.16)
● EST. MAX. SOUND PRESS. LEVEL 92.2 (dBA) (5.1.16) 1B
◎ EST. MAX. SOUND POWER LEVEL (dBA) (5.1.16)

● UTILITY CONDITIONS

ELECTRICITY	VOLTAGE	PHASE	HERTZ
DRIVERS	4160	3	60
HEATER	120	1	60

SYSTEM VOLTAGE DIP ○ 80% ○ OTHER (6.1.5)

STEAM N/A	MAX. PRESS.	MAX. TEMP.	MIN. PRESS.	MIN. TEMP.
DRIVERS				
HEATER				

COOLING TEG: (5.1.19) SOURCE Plant Cooling Glycol System 1B
SUPPLY TEMP. 45 (°C) MAX. RETURN TEMP. (°C) 1B
NORM. PRESS. 2100 (kPag) DESIGN PRESS. 2400 @ 80°C (kPag) 1B
MIN. RET. PRESS. (kPag) MAX. ALLOW. D.P. 70 (kPag) 1B
CHLORIDE CONCENTRATION: (mg / kg)

No.	DATE	REVISION	BY	CHK	ENG.	APPR.	OTHER
1B	31-Jan-08	Re-issued For Purchase	CY	NS	CY	TP	SD
1A	1-Jun-07	Issued For Purchase	GO	ES	CY	TP	SD

DATA SHEET FOR 95G-3215

SUNCOR ENERGY Firebag Project

FOR: Suncor Energy
LOCATION: Firebag Stage 5, Fort McMurray, Alberta
SERVICE: LP BFW Pump
EQUIPMENT NO. 95G-3215

Project No. 03-00061A 1B
DOCUMENT NO. DS95-A-G3215-1 | REV. 1B

NOTES

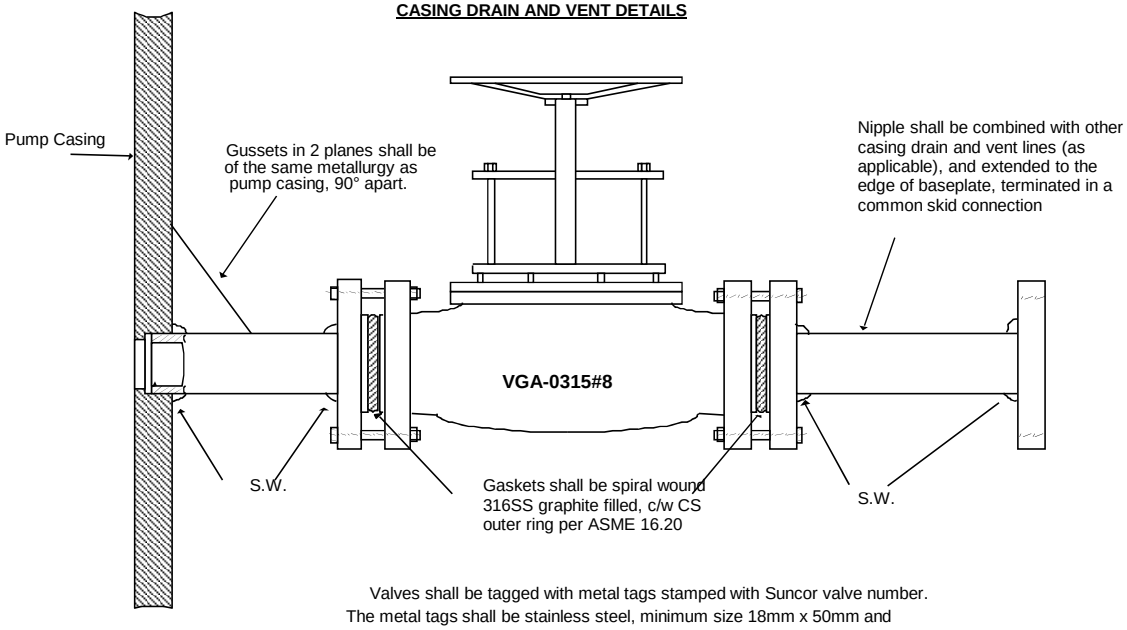
1. Pump lube oil shall be Mobil SCH 626 (ISO VG68)

2. Motor lubrication shall be Mobil SCH 626 (ISO VG68)

3. Seller shall provide casing drains and vents in accordance to the detail below and Suncor Piping Material Class ES on the low point of the pump casing. Seller shall provide a Gate Valve, i.e., Class 300 RF to ASME B16.5, Forged Body w/integral flanges, Bolted Bonned, OS & Y, Solid Wedge, API Trim 8, Integral or Renewable seats, Regular Port, Design to API 602. Seller shall attach a Suncor valve tag number of VGA0315#8. 1B

All casing drains shall be terminated at one common skid edge flanged connection. All casing vents shall be terminated at one common skid edge flanged connection. Each casing drain and vent line shall have provision for individual isolation valve. 1B 1B

CASING DRAIN AND VENT DETAILS



Valves shall be tagged with metal tags stamped with Suncor valve number. The metal tags shall be stainless steel, minimum size 18mm x 50mm and shall be attached to valves with minimum 20 gauge strand SS wire. 1B

4. NPSHR at normal flow of 545 m3/h is 2.0 m. The NPSHA shown is based on the normal temperature of 85°C. 1B

5. Baseplate for pump and motor shall be engineered and designed for non-grout construction. However, grout holes shall be provided.

6. Content for this Data Sheet is based on Suncor Firebag Stage 3 Data Sheet DS93-A-G3215-1 Rev 1D 1B

7. This pump 95G-3215 is identical to Stage 3 Pump 93G-3215. 1B